

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY CHANGA

B.Tech. EE Sem – VII Examination, 2013-14
EE - 402 Electrical power system IV

Date: 20/05/2014, Tuesday

Time: 1.30 am to 4.30p.m

Maximum Marks: 70

Instructions:

1. Attempt all questions from both sections.
2. Use separate answer book for each sections.
3. Figures to the right indicate full marks.
4. Assume suitable additional data, if necessary

SECTION – I

- Q-1 A. Fig.1 shows a three bus system. The series impedance and shunt admittance of each line is $0.026+j0.11$ pu and $+j0.04$ pu respectively. The bus specification and power input etc. at the buses are as under. Determine the values of load angles of bus 2 and 3 and voltage of bus 3 at the end of first iteration by using either NR method.

Bus	P_i	Q_i	P_D (pu)	Q_D (pu)	Bus voltage
1(Slack)	----	----	1	0.4	$V=1.03$ pu $\delta=0$ degree
2 (PV bus)	1.6	----	----	----	$V=1.03$
3 (PQ bus)	---	----	1.2	0.5	----

[14]

- Q-2 A. Compare GS, NR and FDLF methods for load flow analysis.

[10]

- B. Define the accuracy of the load flow method.

[4]

OR

- Q-2 A. Explain the algorithmic steps of Lamda Iterative method.

[6]

- B. Define economic load dispatch and derive exact coordination equation.

[8]

- Q-3 Explain the classification of buses.

[7]

SECTION – II

- Q-4 A. Explain fault analysis [8]
B. Explain type 1 and 2 modifications for fault analysis. [6]
- Q-5 A. Explain turbine speed governing system to control frequency of the system. [8]
B. Explain Series and shunt compensation in transmission system with necessary phasor diagram. [6]
- OR
- Q-5 A. Derive Wave equation of Traveling wave on transmission line. [8]
B. Explain mechanism of lightning stroke. [6]
- Q-6 Discuss different types of protection against surges. [7]
